

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 3: TISSUES IN ACTION****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH03****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Which meristematic tissue is situated at the growing tips of stems and roots and causes an increase in their length?**

- (A) Lateral meristem (B) Apical meristem
(C) Intercalary meristem (D) Vascular cambium

Q.2 The increase in the girth (diameter/thickness) of a tree trunk is brought about by the activity of:

- (A) Apical meristem (B) Lateral meristem (cambium)
(C) Intercalary meristem (D) Parenchyma cells

Q.3 Which simple permanent plant tissue provides both mechanical support and flexibility, allowing stems and tendrils to bend without breaking?

- (A) Parenchyma (B) Collenchyma
(C) Sclerenchyma (D) Xylem vessels

Q.4 The husk of a coconut is composed of which dead, heavily lignified plant tissue?

- (A) Aerenchyma (B) Collenchyma
(C) Sclerenchyma fibres (D) Chlorenchyma

Q.5 Which of the following complex tissue elements in xylem is living and functions to store food materials?

- (A) Tracheids (B) Xylem vessels
(C) Xylem parenchyma (D) Xylem fibres

Q.6 Which phloem component lacks a nucleus at maturity yet remains living and functions in close association with companion cells?

- (A) Phloem parenchyma (B) Sieve tube elements
(C) Bast fibres (D) Xylem tracheids

Q.7 The chemical substance deposited in the cell walls of cork (bark) that makes them impervious to gases and water is:

- | | |
|------------|-------------|
| (A) Lignin | (B) Suberin |
| (C) Pectin | (D) Cutin |

Q.8 Which epithelial tissue lines the alveoli of lungs and blood capillaries to facilitate rapid diffusion of gases and liquids?

- | | |
|---------------------------------------|--------------------------------|
| (A) Stratified keratinized epithelium | (B) Simple squamous epithelium |
| (C) Ciliated columnar epithelium | (D) Cuboidal epithelium |

Q.9 Ciliated columnar epithelium is characteristically found lining which part of the human body?

- | | |
|---------------------|---|
| (A) Skin surface | (B) Respiratory tract (trachea and bronchi) |
| (C) Urinary bladder | (D) Stomach lining |

Q.10 Which connective tissue connects a muscle to a bone and exhibits great fibrous strength with limited flexibility?

- | | |
|---------------|--------------------|
| (A) Ligament | (B) Tendon |
| (C) Cartilage | (D) Areolar tissue |

Q.11 The matrix of bone tissue is dense and hard primarily due to deposits of compounds of:

- | | |
|--------------------------|----------------------------|
| (A) Sodium and Potassium | (B) Calcium and Phosphorus |
| (C) Iron and Magnesium | (D) Sulphur and Silicon |

Q.12 Which specialized connective tissue acts as a thermal insulator and is located beneath the skin and around vital organs like kidneys?

- | | |
|-----------------------|--------------------------|
| (A) Adipose tissue | (B) Areolar tissue |
| (C) Hyaline cartilage | (D) Dense fibrous tissue |

Q.13 Smooth muscle fibres (unstriated muscles) are structurally:

- | | |
|---------------------------------------|---|
| (A) Cylindrical, syncytial, voluntary | (B) Spindle-shaped, uninucleated, involuntary |
| (C) Branched, uninucleated, striated | (D) Multinucleated, branched, voluntary |

Q.14 Cardiac muscle tissue is characterized by which unique combination of features?

- | | |
|--|--|
| (A) Voluntary, unbranched, non-striated | (B) Involuntary, branched, uninucleated, with intercalated discs |
| (C) Voluntary, multinucleated, cylindrical | (D) Involuntary, unbranched, multinucleated |

Q.15 The long, slender cylindrical projection that conducts electrical impulses away from the cell body (cyton) of a neuron is the:

- | | |
|--------------|-----------------|
| (A) Dendrite | (B) Axon |
| (C) Synapse | (D) Myelin node |

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): Meristematic cells have dense cytoplasm, prominent nuclei, and thin walls, but lack vacuoles.

Reason (R): Meristematic cells are actively dividing and do not need to store food or maintain large hydrostatic turgor pressure.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Blood is classified as a fluid connective tissue.

Reason (R): Blood has a liquid extracellular matrix called plasma in which RBCs, WBCs, and platelets are suspended.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Skeletal muscles are called voluntary muscles.

Reason (R): Their contractions can be consciously initiated and controlled at will by our nervous system.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Xylem transports food manufactured in leaves to all other plant parts.

Reason (R): Phloem transports water and mineral salts absorbed by roots upward to the leaves.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Ligaments connect two bones together at a joint.

Reason (R): Ligaments contain very little matrix and are composed of highly flexible elastic fibres.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Microscopic Examination of Plant Stems

During a laboratory session, a student took a thin transverse section (TS) of a herbaceous dicot stem, stained it with safranin and fast green, and mounted it on a glass slide. Under the compound microscope, she noticed: (i) an outer protective epidermis with stomata, (ii) a cortex region beneath containing loosely packed cells with thin cellulose walls and large intercellular spaces, (iii) patches of cells at the corners with uneven pectin thickening, and (iv) vascular bundles containing thick-walled tube-like vessels arranged in rings.

Q.21 The loosely packed cells in the cortex with thin cellulose walls and large intercellular spaces belong to:

- (A) Sclerenchyma
 (B) Parenchyma
 (C) Collenchyma
 (D) Cork

Q.22 If the parenchyma cells in a green stem develop chlorophyll and carry out photosynthesis, they are specifically termed:

- (A) Aerenchyma
 (B) Chlorenchyma
 (C) Collenchyma
 (D) Sclerenchyma

Q.23 The patches of cells showing uneven thickening at corners with no intercellular spaces are identified as:

- (A) Collenchyma (B) Xylem vessels
(C) Sclerenchyma fibres (D) Glandular tissue

Q.24 Why were the xylem vessel walls strongly stained red with safranin?

- (A) Because they contain starch (B) Because safranin selectively stains lignified dead secondary cell walls
(C) Because xylem contains hemoglobin (D) Because xylem vessels are living cells

Q.25 Which specialized epidermal structures observed by the student regulate transpiration and gaseous exchange in the stem?

- (A) Trichomes (B) Stomata with kidney-shaped guard cells
(C) Lenticels only (D) Cuticle layer without pores

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	Apical meristem is present at growing root and shoot apices and brings about primary elongation growth.
Q.2	(B)	Lateral meristems, including vascular cambium and cork cambium, produce secondary growth, increasing plant girth.
Q.3	(B)	Collenchyma has localized pectin thickening at cell corners, giving tensile strength and flexibility without brittleness.
Q.4	(C)	Sclerenchyma consists of long, narrow, dead cells with thick lignified walls that give hardness and rigidity to coconut husk.
Q.5	(C)	In xylem, tracheids, vessels, and xylem fibres are dead; only xylem parenchyma consists of living cells.
Q.6	(B)	Mature sieve tube elements lack a nucleus to facilitate sap transport; their metabolic functions are governed by adjacent nucleated companion cells.
Q.7	(B)	Suberin is a waxy, waterproof lipid deposited in walls of cork cells, making bark impervious to moisture and pathogens.
Q.8	(B)	Simple squamous epithelium consists of extremely thin, flat, tile-like cells ideal for rapid diffusion across barriers.
Q.9	(B)	Cilia on columnar cells beat rhythmically in the respiratory tract to propel mucus and trapped dust particles upward.
Q.10	(B)	Tendons connect skeletal muscle to bone and are tough collagenous cords. (Ligaments connect bone to bone).
Q.11	(B)	Bone matrix is composed of collagen fibers impregnated with hydroxyapatite crystals (calcium and phosphorus salts).
Q.12	(A)	Adipose tissue stores fat droplets in adipocytes, providing cushioning and thermal insulation against cold.
Q.13	(B)	Smooth muscles are unbranched, spindle-shaped (fusiform), with a single central nucleus, working involuntarily (e.g., alimentary canal).
Q.14	(B)	Cardiac muscles are cylindrical, branched, uninucleated, striated, and involuntary, beating rhythmically throughout life without fatigue.
Q.15	(B)	Dendrites receive incoming signals and transmit them to the cyton; the axon carries the action potential away toward the synaptic terminal.
Q.16	(A)	Because meristematic cells divide continuously, storing nutrients or maintaining large central sap vacuoles would hinder rapid mitotic spindle formation.
Q.17	(A)	Connective tissues are characterized by cells distributed within an extracellular matrix. In blood, the fluid matrix is plasma.
Q.18	(A)	Skeletal muscles are attached to the skeleton and are under voluntary somatic nervous control.
Q.19	(D)	Both statements are reversed: Xylem transports water and minerals unidirectionally; Phloem translocates photosynthesized sugars bidirectionally.
Q.20	(A)	Ligaments are strong, elastic connective tissues binding bone to bone to stabilize joints while permitting motion.
Q.21	(B)	Parenchyma is the ground tissue made of living cells with thin cellulose walls and prominent intercellular spaces.
Q.22	(B)	Parenchyma containing chloroplasts capable of photosynthesis is called chlorenchyma.
Q.23	(A)	Localized cellulose and pectin deposition at corner junctions without intercellular spaces is the hallmark of collenchyma.
Q.24	(B)	Safranin binds to lignin, staining the lignified walls of xylem vessels and sclerenchyma bright red.

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.25		Stomata, regulated by two guard cells, open and close to mediate water vapor transpiration and gas exchange (O_2 / CO_2).